

COMPUTER NETWORKS • SDN • AI

AI-Driven Safe Dynamic Network Slicing

for SLA-Preserving SDN Networks

Lightweight Contextual Bandit-Based Dynamic Bandwidth Allocation



PRESENTED BY

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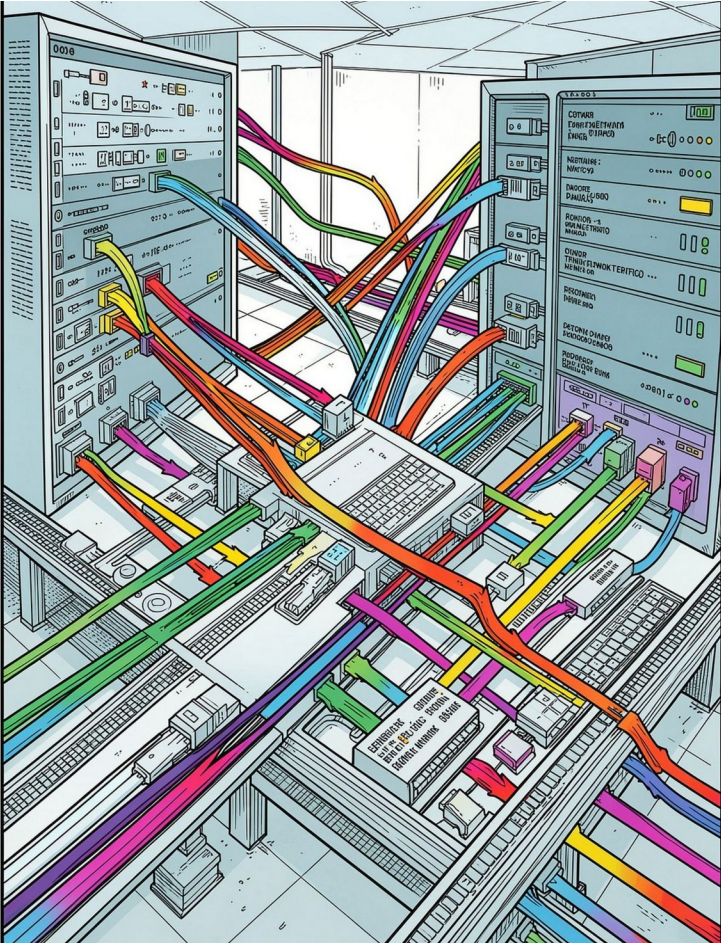
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Guide: Sasikala R • Department: SCOPE

Problem Domain & Motivation

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COMPUTER NETWORKS

SDN

AI

NETWORK SLICING

URLLC

Ultra-reliable, low-latency traffic

Latency-sensitive

< 15 ms guardrail

eMBB

Enhanced mobile broadband traffic

Throughput-heavy

Adapts to demand

Best-Effort

Uses bandwidth left after priority slices

Flexible

Lowest priority

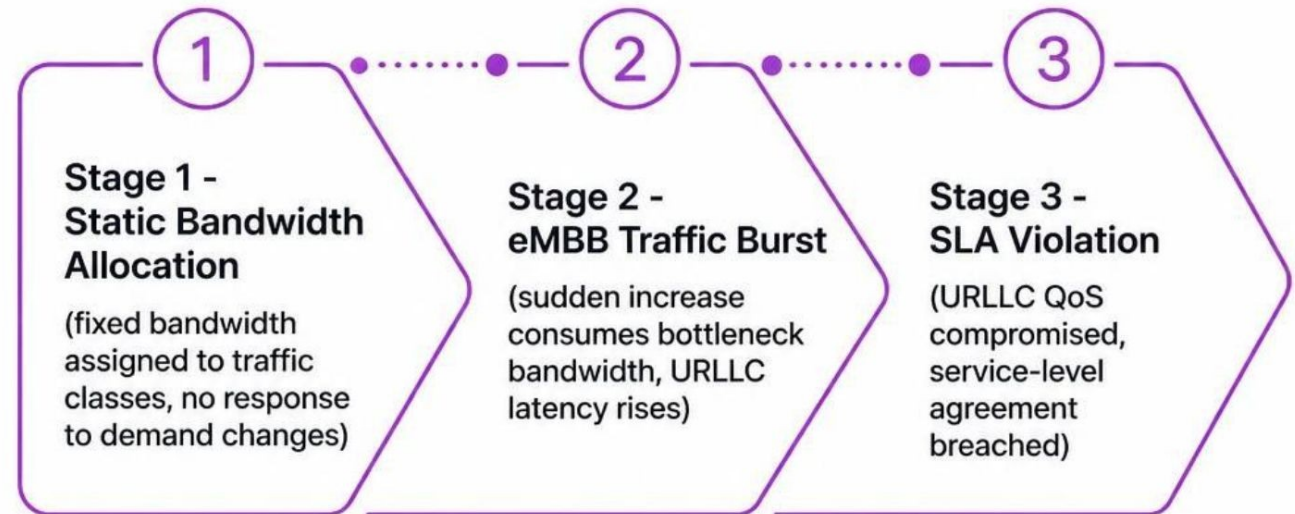
Why it matters: static bandwidth allocation cannot react efficiently when traffic demand changes.

Problem Statement

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What breaks today

- Fixed bandwidth assignments do not respond to changing network demand.
- A sudden eMBB traffic burst can consume bottleneck capacity and increase URLLC latency.
- Many AI approaches rely on complex Deep Reinforcement Learning, adding training time and compute overhead.



Problem: develop a lightweight, AI-driven slicing mechanism that adapts bandwidth in real time while protecting URLLC latency through SLA-aware control.

Literature Survey & Existing System Study

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Method	Approach	Limitation
Koo et al. (2019)	Deep RL	High training complexity
Wei et al. (2020)	DRL Reconfiguration	Computational overhead
Dayot et al. (2022)	Contextual Bandit + DRL	Provisioning-focused
Safe RL (2023)	Safe Resource Allocation	Higher complexity

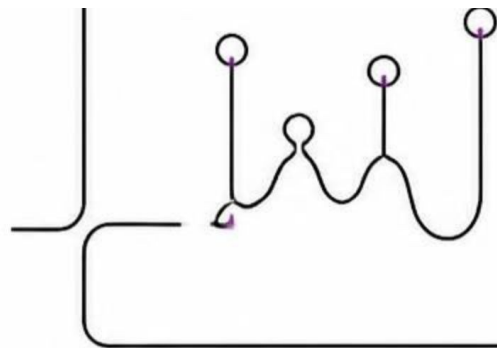
EXISTING SYSTEM

Static network slicing uses predefined bandwidth and fixed queue allocation. It is simple, but cannot adapt to traffic bursts or dynamically optimize resource utilization.

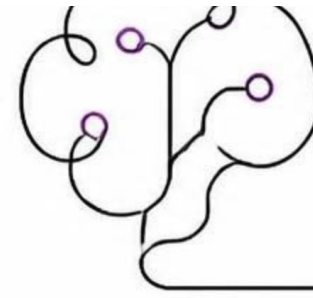
Research Gap

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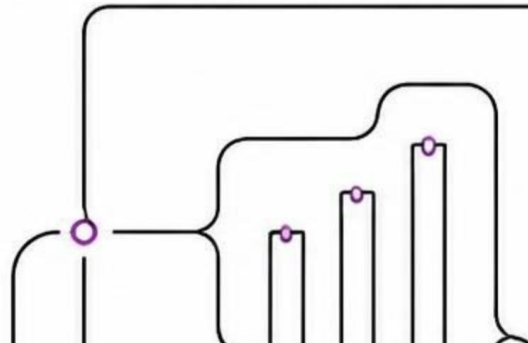
Current approaches usually provide either static allocation or computationally intensive intelligent optimization.



**Real-Time
SDN Telemetry**
(collecting live
network state)



**Lightweight
AI Decision**
(fast contextual
bandit)



**Dynamic
Bandwidth
Allocation**
(adaptive resource



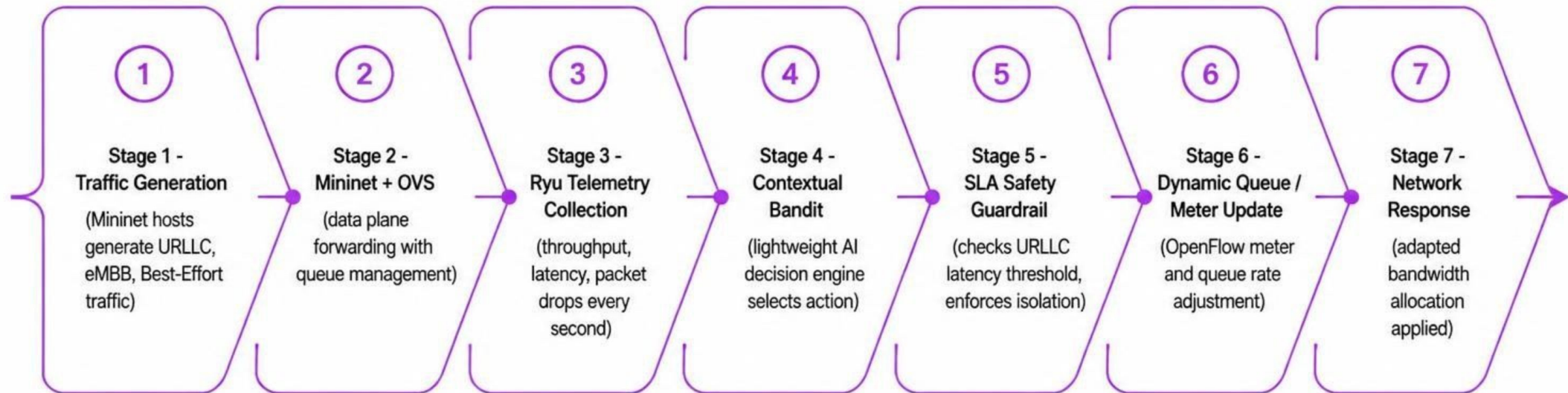
**Deterministic
SLA Protection**
(URLLC latency
auadrail)

Research gap: a lightweight, fast and safety-aware SDN slicing mechanism is needed to allocate bandwidth dynamically without compromising URLLC QoS.

Proposed Solution

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Mininet + Open vSwitch + Ryu Controller + Contextual Bandit decision engine



Traffic slices

URLLC

eMBB

Best-Effort

Decision loop: observe → choose → guard → apply → measure again

Objectives & Novelty

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OBJECTIVES

What we will build

- Implement SDN network slicing using Mininet and OVS.
- Collect real-time throughput, latency and packet-drop telemetry.
- Develop lightweight Contextual Bandit resource allocation.
- Protect URLLC using an SLA guardrail.
- Compare static and dynamic slicing performance.

NOVELTY

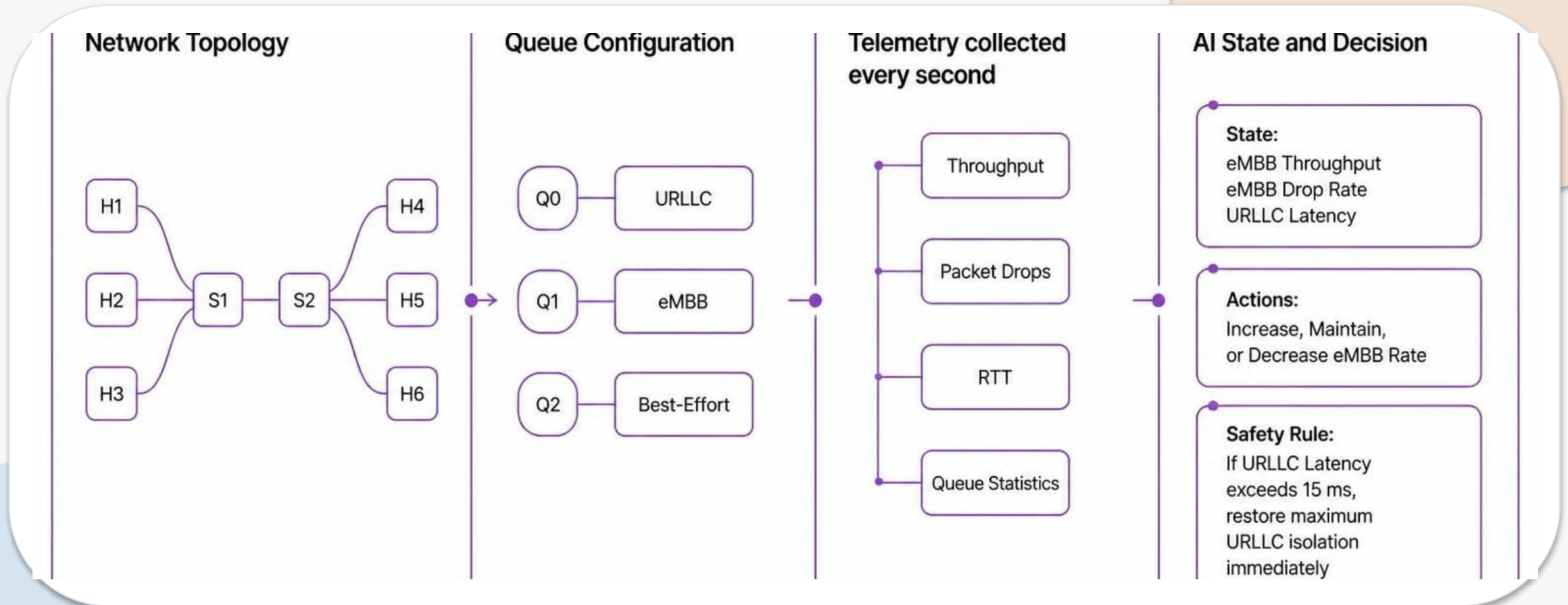
Why this approach is different

- Lightweight Contextual Bandit instead of complex DRL.
- Real-time telemetry-driven bandwidth adaptation.
- Deterministic SLA guardrail for URLLC protection.
- Dynamic meter and queue control in SDN.

Lightweight AI + deterministic safety

Methodology

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SLA safety rule

If URLLC latency exceeds 15 ms, restore maximum URLLC isolation immediately.

Tools, Expected Outcomes & Work Plan

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Tools & Technologies

Python Mininet Ryu Open vSwitch OpenFlow 1.3
iperf3 CSV Matplotlib

Expected Outcomes

- Dynamic bandwidth adaptation
- URLLC latency protection
- Real-time telemetry monitoring
- Performance comparison using latency, throughput and packet drops

6-Week Plan

- Week 1-2 Data Plane Setup
- Week 3 Telemetry Pipeline
- Week 4 AI Engine & Guardrail
- Week 5 Benchmark Testing
- Week 6 Analysis & Documentation

References

- 01 Koo et al., Deep Reinforcement Learning for Network Slicing, 2019.
- 02 Wei et al., Dynamic Network Slice Reconfiguration using DRL, 2020.
- 03 Dayot et al., Deep Contextual Bandit-Based Slice Provisioning, 2022.
- 04 Safe and Fast Reinforcement Learning for Network Slicing Resource Allocation, 2023.
- 05 Xie et al., DRL-Based Resource Allocation for Network Slicing, 2022.

THANK YOU

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